QPath — Supplementary Materials

Companion to **"QPath: An Instrumented Generative-AI Platform for Introductory Quantum Programming Education with Qiskit"**

Anonymous reviewer-oriented supplementary document | Version 1.2 | 12 November 2026

This supplement documents the original pilot cohort, assessment item stems, export schema, descriptive analyses, LLM response ratings, and reproducibility procedure. It is designed to accompany a six-page IEEE-format manuscript; it is **not** an additional experiment or a validation of teaching effectiveness. No author identifiers or participant-level records are included.

Evidence note. Figures, methods, and interpretation refer to the companion manuscript. Quantities marked "recomputed" were independently checked against the 7 November 2026 research export; operational prompt wording, complete test distractors, and the original consent wording were not available in that export.

S1. Pilot cohort, protocol, and reconstruction

The original study used one arm with no contemporaneous control. Learners registered, affirmed electronic study consent, completed a pre-test, accessed the guided Qiskit learning path and optional LLM coaching, then completed a post-test and survey. A complete case required consent, a pre-test, at least one lesson activity, at least one AI interaction, a post-test and survey completion. The flow below reports records, not randomized allocation.

Milestone	Count	Interpretation
Accounts registered by the cutoff	24	Researcher-reconstructed registration cohort
Pre-tests submitted	21	87.5% of registered accounts
Post-tests submitted	8	All eight also completed the survey
Pre-test without post-test	13	Other pre-test completers; not necessarily zero lesson exposure
Complete pre/post/survey cases	8	38.1% of the 21 pre-test completers
Pre-to-post noncompletion	61.9%	13/21; possible selection bias

Cohort reconstruction used a retrospective time boundary: registration and AI interaction timestamps strictly before 1 July 2026 (i.e., through 30 June). The later 7 November export also contains two additional accounts and five later AI interactions. These later records are excluded from the published original-cohort analyses. The boundary was applied during reconstruction; advance specification is not documented.

The eight complete cases had a baseline pre-test mean of 53.75% (SD 31.14), compared with 45.38% (SD 21.84) among 13 other pre-test completers. This 8.37 percentage-point descriptive difference was not significance-tested. Since complete-case status includes having used AI, the paired outcome estimate may overrepresent engaged learners.

S2. Assessment instruments and survey mapping

The source export contains exactly 10 unique pre-test question stems and 10 unique post-test stems for the original cohort, each with a keyed correct response and a short explanation. It does **not** contain the complete list of multiple-choice distractors as presented to students, and formal form-equivalence validation was not documented. Below are the original exported stems and correct responses, not a newly generated instrument.

Pre-test: ten original item stems

`pre_q1` — Quantum circuit
In Qiskit, what does a QuantumCircuit(1, 1) usually represent?
Key: A circuit with one quantum bit and one classical bit

`pre_q2` — Qubit, state, and measurement
Before any gate is applied, a newly created qubit in a basic circuit is usually initialized to:
Key: |0>

`pre_q3` — Hadamard gate
What is the main role of a Hadamard gate H applied to |0>?
Key: It creates a superposition state

pre_q4 — Measurement
What does measurement do in an introductory quantum circuit?
Key: It maps the quantum state to a classical outcome

pre_q5 — Shots and counts
Why are circuits often executed with many shots?
Key: To estimate the distribution of possible measurement outcomes

pre_q6 — CNOT gate
In a CNOT gate, the target qubit is flipped when:
Key: The control qubit is in state 1

pre_q7 — Classical vs quantum
Which statement best distinguishes a qubit from a classical bit before measurement?
Key: A qubit can be described by amplitudes and may be in superposition

pre_q8 — Qiskit syntax
In Qiskit, which line applies a Hadamard gate to qubit 0?
Key: qc.h(0)

pre_q9 — Entanglement intuition
A common purpose of using H on one qubit followed by CNOT is to demonstrate:
Key: Correlated quantum outcomes

pre_q10 — Debugging
A frequent beginner mistake in Qiskit measurement is:
Key: Measuring into a classical bit that was not allocated

Post-test: ten original item stems

post_q1 — Quantum circuit
What are the two resources specified in QuantumCircuit(2, 2)?
Key: Two qubits and two classical bits

post_q2 — Qubit, state, and measurement
If a qubit remains in |0> and is measured many times, what result is expected?
Key: Mostly 0

post_q3 — Hadamard gate
After applying H to |0> and measuring many shots, the counts should be approximately:
Key: A balance between 0 and 1

post_q4 — Measurement
Why is a classical bit used in qc.measure(0, 0)?
Key: To store the measurement outcome of qubit 0

post_q5 — Shots and counts
If a result dictionary is {'0': 510, '1': 490}, what does it suggest?
Key: An approximate 50/50 outcome distribution

post_q6 — CNOT gate
In a simple Bell-style circuit, why can outcomes 00 and 11 appear more often than 01 and 10?
Key: Because the qubits are correlated by H and CNOT

post_q7 — Classical vs quantum
Why should a learner avoid saying that a superposed qubit is simply 'both classical values at once'?
Key: Because amplitudes and measurement probabilities are more precise than that simplified phrase

post_q8 — Qiskit syntax
Which instruction measures qubit 0 into classical bit 0?
Key: qc.measure(0, 0)

post_q9 — Entanglement intuition
Which pair of operations is commonly used to introduce correlated two-qubit outcomes?
Key: H then CNOT

`post_q10` — **Debugging**
A student writes QuantumCircuit(1, 0) then qc.measure(0, 0). What is the likely issue?
Key: No classical bit was allocated for the measurement result

Assessment scoring: each original pilot form comprised ten items and yielded scores in ten-point percentage increments. Full answer-option sets and test-form administration screens must be retrieved separately from an archived application version for a complete operational instrument.

Learner survey: available variable labels

The export provides seven named 1–5 numerical fields but does not preserve the exact wording shown to participants. Labels below are derived only from field names. Eight survey records belong to the eight complete cases.

Survey variable label	Mean (1–5)
Scaffolding clarity	3.875
Qiskit examples	3.875
AI feedback	3.875
Exercise generation	3.125
Reflection prompts	3.625
Over-reliance awareness	3.375
Usability	3.875

Overall seven-item survey mean = 3.6607/5. Interpret as self-report only.

S3. LLM instruction and prompt evidence

The prototype AI tutor used Llama-3.1-8B-Instant through the Groq API, with support modes including conceptual explanation, guided hints, practice generation, feedback, reflection and a local curated-template fallback. Context variables described in the manuscript include current module, target concept, learner attempt text and requested support type. Concept Builder events are action logs rather than additional LLM model responses. In the original cohort, 14 events consisted of seven `concept_builder_request` entries and seven recorded follow-up actions (three visual cards, one mini-quiz, one misconception check, one analogy, and one simpler explanation). The two `curated_template` output records in the 90-row AI-interaction stream came from the Concept Builder: one mini-quiz and one misconception check (`model=concept-builder-v12.4`). The action log links these two curated responses to their corresponding events; therefore the streams overlap and must not be summed.

Interpreting the support modes and activity features

Explain Simply requests a less technical restatement of a selected concept or explanation, while **Ask Me a Question** requests a guiding or comprehension-check question. Neither mode was evaluated as a separate experimental condition. The exercise-generation feature offers a lesson-context practice prompt rather than a formally validated item bank; seven original-cohort LLM calls carry the exact task label `Generate a practice exercise`. These are request counts, not automatically validated exercises or successful learning outcomes.

The **Local Fallback** is a rule/template path intended for cases in which external generation is unavailable or fails. Its activation flag was zero across all 90 original-period AI interactions. The two Concept Builder outputs classified as `curated_template` must not be misreported as fallback activations. Function availability in the platform architecture is distinct from observed use and pedagogical effectiveness.

Illustrative prompt structure from the main manuscript

> You are a quantum programming tutor. The learner is working on the current module: [module name]. The target concept is: [concept]. The learner attempt is: [learner text]. Provide a short hint or guiding question without giving the final answer directly. If Qiskit code is relevant, connect the explanation to a minimal Qiskit example.

Evidence note. This quoted prompt is illustrative, not a verified byte-for-byte historical prompt. The available research export has prompt-template metadata but intentionally excludes live raw prompts and replies. Exact deployed system-prompt files and prompt versions should only be archived if retrievable and verified from source code.

S4. LPQS scoring, distribution, and uncertainty

One evaluator rated 42 responses from nine learners (five also belonged to the eight complete cases). All rated model replies came from Llama-3.1-8B-Instant. Each response received integer scores from 1 to 5 on seven criteria. LPQS was calculated as the arithmetic mean of the seven ratings for a response, then averaged descriptively over rated responses. This approach does not establish interval-scale measurement validity or inter-rater reliability.

Criterion	Mean	Median	SD (sample)
Conceptual accuracy	3.60	3	0.86
Answer relevance	3.45	3	0.74
Pedagogical clarity	3.24	3	0.48
Scaffolding quality	3.21	3	0.52
Qiskit alignment	3.71	3	0.97
Reflection support	3.17	3	0.44
Personalization	3.19	3	0.51

Rating profile group	N	Mean LPQS	Interpretation
All seven criteria rated 3	27	3.00	64.3% of rated responses
At least one criterion not 3	15	4.03	Descriptive subgroup; not a second rater
All rated responses	42	3.37	Overall median 3.00

The platform interface described in the manuscript has rating sliders initially set to 3. The export cannot distinguish deliberate midpoint judgments from unchanged defaults, nor prove that a full transcript was opened for each historical rating. None of the 42 recorded criterion scores is below 3. The selection of the 42 responses for rating was not probability-based or documented as random. Scores may be correlated within learners. Accordingly, LPQS is a preliminary descriptive rubric application, not a validated measure of tutor quality.

S5. Reproducing the statistical analyses

All calculations below use the eight matched complete cases from the original cohort. Primary analysis is descriptive, and inferential statistics are exploratory due to $n = 8$, selection through completion, the absence of a control condition, and the short unvalidated tests.

Quantity	Computation	Recomputed result
Mean raw paired change	mean(post - pre)	30.00 percentage points
Mean pre-test	8 matched observations	53.75%
Mean post-test	8 matched observations	83.75%
Class-average normalized gain	(83.75 - 53.75) / (100 - 53.75)	0.64865 (reported 0.65)
Paired t-test	df=7, two-sided	t=2.5752, p=0.03673
95% t CI for mean change	30 +/- t(0.975,7) x (SDdiff/sqrt(8))	[2.45, 57.55] pp
Cohen dz	mean difference / SD of differences	0.9105
Approx. Hedges gz	dz x [1 - 3/(4*7-1)]	0.8093
Cohen dav	30 / mean(SDpre, SDpost)	1.3586
Wilcoxon signed-rank	W=2; 7 nonzero pairs; two-sided exact	p=0.046875 (paper: 0.047)
Pre/post Pearson correlation	Pearson r across 8 matched cases	0.066

Both reported p-values are just below 0.05; they do not provide independent confirmation or causal evidence. Two pre-test scores were 100%, one paired change was zero, one was negative, and six were positive. The group-level Hake gain of means is defined despite individual ceiling cases. Reporting the average of individual normalized gains would be inappropriate here because individual gain is undefined when pre-test=100%.

Reproduction levels: the attached code checks published calculations from aggregate summaries. An additional optional `private_audit.py` independently replays the summary directly from the researcher-held original export ZIP, without releasing that ZIP or participant-level records.

S6. Instrumentation and data dictionary

The platform research export is a multi-table workbook with CSV copies. In the 7 November snapshot, the later, evolving platform export contains 26 accounts and 95 AI interactions; the original pilot uses only records within the retrospective window. Particular tables supply distinct evidence types:

Table	Relevant fields / role	Caveat
progress_summary	Registration state, pre/post flags, case completeness	AI interaction count is a later snapshot counter
test_attempts	Attempt type, score, total, keyed response map	No participant code in this table
question_responses	Question stems, keyed correct responses, item scores	Rows link to participants and timestamps; withheld
ai_interactions	Timestamp, mode, provider/model, task, latency	90 original period rows after filtering
llm_evaluations	Seven LPQS criteria; link to interaction time	Some ratings entered after original period
survey_item_scores	Seven 1–5 numerical items	Exact original question wording not in export
consent_records	Consent-version field and recorded timestamp	Does not contain full text of informed consent
ai_request_timing_parsed	Delay before AI request	Only 12 original-period observed timings
learner_attempts	Structured attempts / text if available	Zero rows in provided export
concept_builder_events	Builder action logs	14 original-period events, not 14 AI replies

Event-count reconciliation for the original cohort

Export stream / status	Count	Interpretation
AI interactions, total	90	88 model calls plus two curated-template response records
Concept Builder events	14	Seven request events and seven action events; separate logging stream
Explicit practice-exercise requests	7	LLM calls with task label `Generate a practice exercise`; output correctness untested
Simulator opened	24	Event occurrences, not distinct learners or verified execution
Animation viewed	24	Event occurrences, not verified learning outcomes
Simulator completed	2	Learner-reported activity completion, not a simulator-output audit
Local fallback used	0	No fallback activation among the 90 original-cohort AI interactions
Timed AI requests	12	Only these events support mean time-before-request of 169.3 s

The original-period AI interactions comprise 88 LLM calls and two curated-template outputs (Concept Builder mini-quiz and misconception check), both linked to matching action records. The 14 Concept Builder event logs represent a separate, partly overlapping action stream and must not be added to the 90 interactions. Account-level usage median was 2, with five accounts having zero interaction and a top account contributing 20/90 (22.2%). The timing mean of 169.275 s derives from 12 observations, not the full interaction log.

S7. Ethics, privacy, availability, and release restrictions

The pilot was conducted without referral to an institutional research ethics committee; no formal committee approval or exemption determination was obtained. The platform required electronic study consent at initial registration, and the export includes dated consent-version records (v1). The exact consent-screen wording is not included in the available export. An institutional review of the ethical status, informed-consent scope, and permitted data-sharing terms remains appropriate before a public data release. No unverified exemption is claimed.

The strict research export excludes direct names and high-risk raw free text, but retained stable pseudonymous codes, event times, and linkable observations in the researcher-held ZIP. These are not equivalent to irreversibly anonymous public data. The anonymous repository currently supplies the supplementary PDF with aggregate statistics, question stems and keys (but no personal answers), and instrument variable labels. The platform source, executable replay code, and participant-level export are not hosted there. The item stems/keys are reviewer-only material and may require separate permission before public posting or reuse in teaching.

Excluded from public or anonymous review artifacts: raw participant rows, per-person test scores and ratings, stable participant codes, operational transcripts, specific timestamps, participant-code crosswalk, database URLs, provider keys, raw consent records, and identifiable feedback. Authorship identifiers must also be withheld if anonymous review applies.

The ISPA 2026 webpage specifies a six-page IEEE-format PDF for the main manuscript but does not unambiguously authorize supplementary uploads. Confirm that CMT supports additional files before submitting this supplement. Public DOI publication should follow, not precede, an anonymity and privacy review.

Reproduction instructions — Files and execution modes

Public aggregate-only mode: after extracting the reproducibility package, install `code/requirements.txt` and run `python code/reproduce_aggregates.py`. It checks class-average normalized gain, paired t, confidence interval, effect sizes, and the conventional signed-rank tail probability from stored sufficient summaries. It does not regenerate primary source events.

Private source replay: the researcher may run `python code/private_audit.py -source-zip` locally, followed by the local path to the held export ZIP (for example, `/private/path/to/3alimnia_reproducibility_bundle_20261107_102609.zip`). This reconstructs the period-specific cohort, counts, test statistics, LPQS, and AI use from the private export; it emits only aggregates. Do not upload the source ZIP to a public repository.

Expected outcomes: $n = 24$ accounts, 21 pre-tests, 8 post-tests, 90 original AI interactions, 42 rated LLM responses, normalized gain 0.64865, paired t $p = 0.03673$, Wilcoxon $p = 0.046875$, and LPQS mean 3.36738. The separate local audit script was tested against the provided private export; it passed all critical checks.

Source and status: Calculations derive from the QPath manuscript and the 7 November 2026 research export/codebook; the gain-of-means definition follows Hake (1998). Event counts in S3/S6 were verified against the cutoff-filtered export; interface-mode descriptions follow the manuscript and were not independently established by the exported rows. No new experiments or raters were introduced.